

Bhargav Borah

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EDUCATION

- **Texas A&M University** College Station, TX
Master of Science in Computer Engineering; GPA: 4.0/4.0 | Advisor: Dr. Guni Sharon Aug 2025 – Present
- **National Institute of Technology** New Delhi, India
Bachelor of Technology in Electrical and Electronics; CGPA: 7.84/10.00 Dec 2021 – July 2025

RESEARCH

- **Spectral Prior-based Representation Shaping for Offline Reinforcement Learning**
Borah and Sharon *Under Submission, 2026*

EXPERIENCE

- **Chang Gung University** Taoyuan, Taiwan
Research Intern, Artificial Intelligence and Big Data Computing Lab (Prof. Prasan Sahoo) Dec 2024 – June 2025
 - **Attention Mechanism Research:** Extended the standard transformer attention mechanism to operate over hierarchical segments alongside individual tokens, enhancing long-range contextual modeling.
 - **Theoretical Contribution:** Formulated a segment-based contextual aggregation framework to expand the effective receptive field in long-sequence transformer models.
 - **Implementation:** Built and integrated custom PyTorch modules enabling systematic experimentation with hybrid token-segment attention architectures.
- **Indian Institute of Technology, Patna** Patna, India
Research Intern (Prof. Rishav Singh) June 2024 – July 2024
 - **Hybrid Architecture Design:** Designed a novel hybrid architecture fusing convolutional neural networks with Kolmogorov–Arnold Networks (KANs) to improve non-linear feature representation.
 - **Medical Imaging Application:** Applied the CNN-KAN hybrid to automated detection and classification of degenerative spine conditions from lumbar MRI scans, advancing diagnostic accuracy.
 - **Modeling & Training:** Engineered custom PyTorch modules for seamless KAN integration and fine-tuned training pipelines tailored to the constraints of medical imaging datasets.

PROJECTS

- **MCTS for LLM Mathematical Reasoning (CSCE 642):**
 - Designed a reasoning framework combining Monte Carlo Tree Search with Tree-of-Thought-style structured exploration and process reward signals to enhance step-by-step mathematical problem solving in LLMs.
 - Trained a lightweight policy network via experience replay on collected MCTS trajectories, using a fine-tuned DeepSeek-R1-Distill-Qwen-1.5B backbone with symbolic verification and log-likelihood intermediate rewards.
- **Circuit Development During Fine-Tuning (ECEN 649):**
 - Applied mechanistic interpretability techniques—path-patching, Desiderata-based Component Masking (DCM), and Cross-Model Activation Patching (CMAP)—to track how circuits evolve across LoRA, QLoRA, and full fine-tuning checkpoints.
 - Demonstrated that parameter-efficient fine-tuning preserves the base model’s circuit structure while progressively enhancing component functionality, with QLoRA exhibiting more gradual but comparable circuit development to LoRA.
- **PhysicsPDEAgent: Agentic AI for PDE Solver Generation (ECEN 744):**
 - Built an agentic LLM framework that automatically generates, verifies, and iteratively repairs numerical PDE solvers by embedding physics constraints—conservation laws, stability conditions, energy bounds, and positivity requirements—into the generation and correction loops.
 - Evaluated across four PDE families (advection, Burgers’, Darcy flow, reaction-diffusion) using Claude Sonnet 4.5, Gemini 2.5 Flash, and Gemma-3-27B-it, producing physically consistent solvers with a hierarchical physics verification system.
- **Continuous Thought Machine on MNIST:**

- Reproduced the *Continuous Thought Machines* (Darlow et al., 2025) architecture, which models neural synchronization and an internal temporal dimension inspired by biological cognition.
- Validated the implementation on MNIST, successfully replicating the paper’s reported accuracy and prediction behavior.
- **Vision Transformer for Image Classification:**
 - Implemented a Vision Transformer (ViT) from scratch following *An Image is Worth 16×16 Words* (Dosovitskiy et al., 2020), including patch embedding and multi-head self-attention.
 - Evaluated classification performance on CIFAR-10, benchmarking against CNN-based baselines to assess architectural tradeoffs.
- **Neural Machine Translation with Transformers:**
 - Constructed an English–German neural machine translation system from scratch following *Attention Is All You Need* (Vaswani et al., 2017).
 - Implemented multi-head self-attention, positional encodings, and a full encoder–decoder pipeline with training and inference support.
- **Neural Style Transfer for Artistic Rendering:**
 - Faithfully reproduced *A Neural Algorithm of Artistic Style* (Gatys et al., 2015) in PyTorch, leveraging VGG feature representations for style and content separation.
 - Generated high-fidelity artistic renderings by transferring texture and style from reference artworks onto arbitrary content images.

RELEVANT SKILLS

- **Programming:** Python, PyTorch, JAX, vLLM, Unsloth, HuggingFace, LangChain, LangGraph, SQL

AWARDS AND ACHIEVEMENTS

- Won a silver medal for ranking 104/3325 as part of a team in the Kaggle competition ‘CIBMTR - Equity in postHCT Survival Predictions’

RELEVANT COURSEWORK

- ECEN 649 – Pattern Recognition
- CSCE 642 – Reinforcement Learning
- CSCE 625 – Artificial Intelligence
- ECEN 744 – Scientific Machine Learning
- CSCE 753 – Computer Vision and Robot Perception
- CSCE 689 – Advanced Topics in Deep Reinforcement Learning